David Reyna
Aircraft Design

Personal Jet

Introduction

This aircraft will allow for a small family to travel around. My product will fill the need of people who want an affordable and efficient personal airplane. For people who are tired of waiting at airports or all the strict regulations on luggage. This model will be the Honda Civic or Toyota Camry of aviation. Whether it be to sight-see around a city or travel to a different state, the personal aircraft will have various applications.

Mission and Top Level Vehicle Design Requirements.

The mission of the aircraft starts on the ground, where it will warm up and run diagnostics to make sure the plane is ready for flight.. Next it will take off on a short runway. It will then climb up to a desirable altitude and stay within that range. While in that range, it will cruise to its destination.Once near the destination it will be able to loiter if needed while waiting for clearance to land. It will then start slowly descending. Next it will perform a smooth and comfortable landing. Once on land again the plane will be able to roll to a desired parking space where it will then run diagnostics to determine the state of the airplane.

In order for the mission to be successful, some design requirements will be as follow:

- Shall be able to withstand rain, snow, hail and similar type weather while in flight.
- Shall have a range of 1000-1200 nautical miles
- Shall comfortably accommodate 4-5 passengers approximately 180lbs each.
- Shall be able to carry an additional payload of 400lb
- Shall be able to run constant background diagnostic to make sure the airplane is in peak condition
- Shall have safety measures to prevent takeoff if it does not pass the preflight diagnostics.
- Shall be operable even for pilots with no prior flying experience.
- Shall be able to takeoff and land within 2000 feet.
- Shall be able to operate within 30000 and 35000 feet.
- Shall be able to cruise at 300mph.
- Shall meet FAA Part 23 landing load and passenger comfort criteria
- Shall be drivable for short distances.
- Shall be maintainable by a single technician using standard ground equipment

Similar Designs

Cirrus Vision Jet SF50-Small single engine jet

- Performance specifications
    - Cruise speed: 300-350 mph
    - Range:1100-1300 nm
    - Altitude: 31000ft
    - Takeoff distance: 2000ft
    - Landing distance: 1600ft
- Weight
    - Max takeoff weight: 6000lb
    - Empty weight: 3500lb
    - Payload:400-600lb
    - Useful load:2500lb
- Capacity
    - 1 pilot
    - 4-6 passengers
- Cost and lifecycle
    - New price:2-2.4 million dollars
    - Operation cost-$600-$800 hourly
    - Life expectancy:12000 flight hours
- Notes
    - Require jet training

Eclipse 500/550- Twin engine entry-level business jet

- Performance specifications
    - Cruise speed: 430 mph
    - Range:1150nm

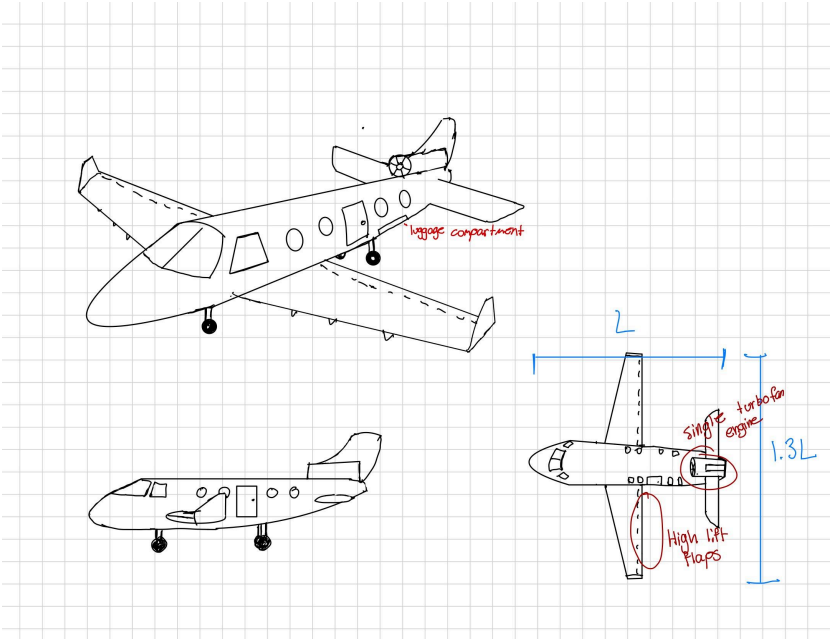
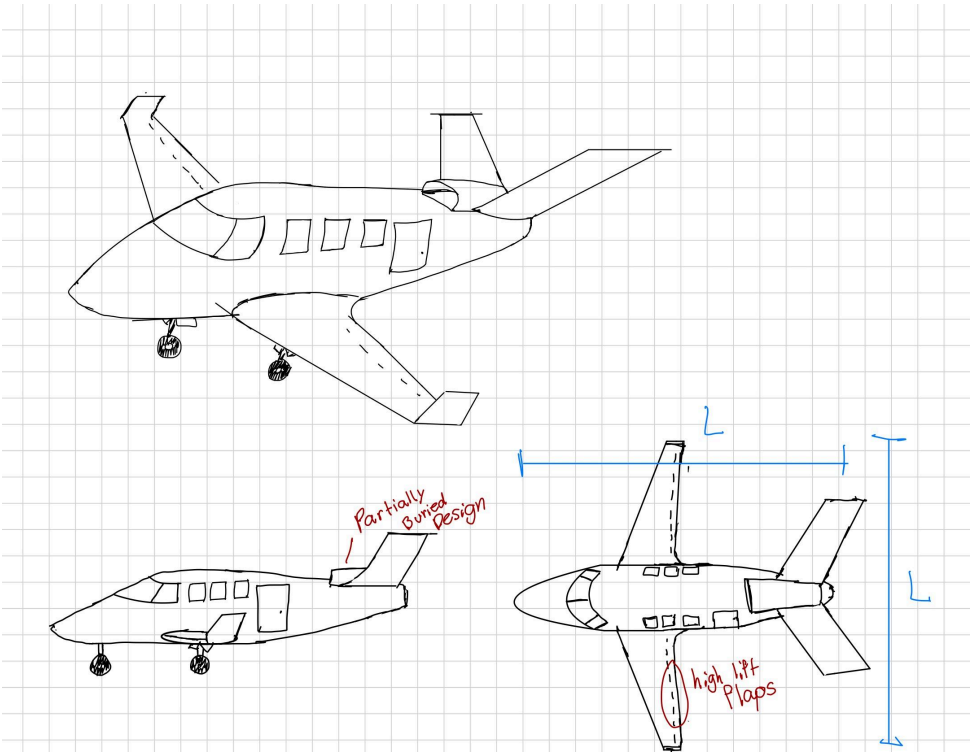
    - Altitude: 41000ft
    - Takeoff distance: 2,155–2,433 ft
    - Landing distance: 2,040–2,790 ft
- Weight
    - Max takeoff weight: 6000lb
    - Empty weight: 3540-3600lb
    - Payload:700-1100lb
    - Useful load:2200-2400lb
- Capacity
    - 2 pilots
    - 4-5 passengers
- Cost and lifecycle
    - New price:1.3 million dollars
    - Operation cost-higher than single engine
    - Life expectancy:20000 flight hours
- Notes
    - Requires jet and multi-engine training

Critical Performance Attributes

Some key attributes it must achieve in order for the mission to succeed would be:

- Range with max payload: no less than 1100 nm
- Cruise speed: 300mph
- Fuel efficient:Target fuel consumption of approximately 120–180 gallons per hour.
- Maximum operating altitude of 35000ft.

Conceptual sketches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